THIRUMARAN DEEPAK

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PROFILE

Incoming University of Washington student and software developer focused on robotics, reliable AI systems, and real-time infrastructure. Built production tools across Android security, LLM-controlled media pipelines, and hardware-software integration, with experience owning projects from prototyping through deployment.

TECHNICAL SKILLS

Languages Python, JavaScript, Java, C++, C#, SQL, HTML/CSS

Frameworks & Web React, Node.js, FastAPI, Vite, Tailwind

AI & Infrastructure WhisperX, Gemini API, FFmpeg, MoviePy, WebSockets, PathPlanner

Tools Git, Azure Cloud, Microsoft Dynamics 365

EXPERIENCE

University of Washington, Department of Electrical & Computer Engineering | *Undergraduate Researcher* Aug 2026 – Present

- Conducting reinforcement learning research for robotic manipulation with PhD researcher Jake Gonzales, within research advised by Prof. Lillian Ratliff and Prof. Behçet Açıkmeşe, focused on learning and control for robotic systems.
- Training and evaluating PPO-based policies for Franka robotic manipulators in MuJoCo Playground, beginning with pick-and-place and block-stacking benchmarks.
- Building the simulation and experimentation pipeline for robotic policy training, including environment setup, reward design, training runs, evaluation, and performance analysis.

Microsoft AI | *Copilot Student Ambassador* Aug 2026 – Present

- Represent Microsoft Copilot across the University of Washington campus, leading student workshops, campus events, and hands-on demonstrations of Copilot features and workflows.
- Provide direct feedback to the Copilot team based on how students use the product in coursework and projects.

Canary Technologies | *Applied AI Intern* Oct 2025 – Jun 2026

- Architected an LLM agent layer exposing video editing operations (trim, concatenate, subtitle, mix) as callable tools, enabling natural-language control of the render pipeline with real-time WebSocket progress streaming.
- Built an FFmpeg-first render pipeline with MoviePy for frame-level effects, handling 500MB+ disk streams to keep renders fast and memory-stable across multiple aspect ratios.
- Designed a human-in-the-loop review UI with editable transcripts and override controls, ensuring the AI agent's output matched user intent before rendering.

Ramen Robotics, FRC Team 9036 | *Software Engineer & Tester* May 2025 – Jun 2026

- Built and tested autonomous routines in PathPlanner — authoring paths, sequencing commands, and re-tuning routes against measured robot performance across the season.
- Developed and debugged the robot control software at practices and competitions under match-day time pressure; prototyped a vision system for camera-based targeting.
- Partnered with the mechanical, electrical, and drive teams to translate match strategy into reliable, repeatable robot behavior.

Google | *Android Security Intern* Oct 2024 – Jun 2025

- Year-long internship with Google's Android Security team (Kirkland) under lead mentor Greg Wroblewski; chose and owned two independent projects end-to-end.
- Built an Android fuzzer that targets new OS builds to surface crashes, capturing each with its ID and stack trace, then uses the Gemini API to generate plain-language root-cause explanations for each bug.
- Built a security bulletin scraper that ingests the Android Security Bulletin as a labeled reference set, uses Gemini to classify newly found bugs against it, and publishes per-bug summaries back to the bulletin board.

Lakehills Orthodontics | *Healthcare Systems Intern* Sep 2023 – Jun 2024

- Designed a patient progress-tracking system in Dynamics 365 to replace a mix of spreadsheets and paper records, standardizing data collection across staff and cutting workflow delays by 25%.
- Delivered the rollout two weeks early and audited 50+ patient records during the transition, improving data reliability by 40%.

Quadrant Technologies | *App Development Intern* Nov 2022 – Jun 2023

- Built a Python/JavaScript patient data management app that cut staff retrieval time by 30%, with a data layer built to absorb inconsistent input formats from across the facility.
- Kept the build aligned with healthcare compliance requirements, tracking budget against milestones and finishing 15% under the original estimate.

EDUCATION

University of Washington | *B.S., Computer Science* — Tacoma, WA · Expected Jun 2030

Bellevue College (Running Start) | *Computer Science* — Bellevue, WA · Graduated Jun 2026

Bellevue Big Picture High School | *High School Diploma* — Bellevue, WA · Jun 2026

LEADERSHIP & ACTIVITIES

Coding Club (Co-Leader) · Newspaper Club (Co-Leader, Article Editor) · Medicine Club (Co-Leader) · Salvation Army (Volunteer)